

**CANDIDATE
NAME**

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8872/02
22 August 2013
2 hours

Additional Materials: Data Booklet
 Writing Papers

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.

Answer **all** questions in the space provided in the booklet.

Answer any **two** questions on separate answer papers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

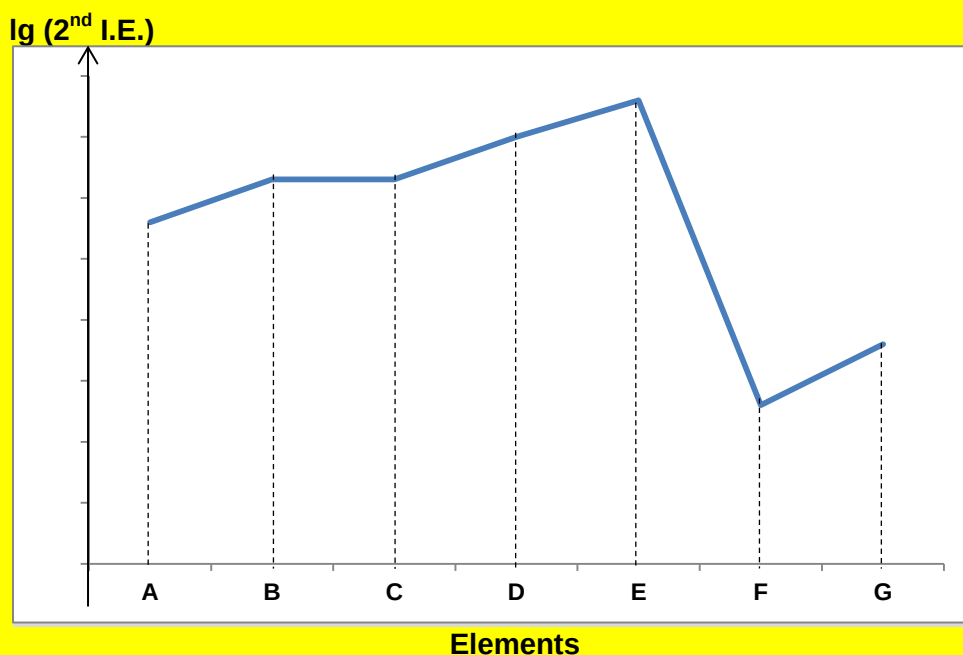
For Examiner's Use				
MCQ P1				
Section A	1	2	3	Total
B4				
B5				
B6				
Total	/80			

Section A

Answer **all** the questions in this section in the spaces provided.

1	(a)	Arsenic acid, H_3AsO_4 , is a colourless acid that is used as a wood preservative and as a reagent in the synthesis of dyestuff. It is slowly formed when elemental arsenic, As, is treated with ozone in the presence of moisture. Oxygen gas is known to be a side product for the reaction.	For Examiner's use
	(i)	Write a balanced chemical equation for the above mentioned treatment process.	
		$2\text{As} + 3\text{H}_2\text{O} + 5\text{O}_3 \rightarrow \text{H}_3\text{AsO}_4 + 5\text{O}_2$ [1]	
	(ii)	Arsenic acid can oxidise iodide ions to iodine and itself is reduced to H_3AsO_3 . Write a half-equation for the reduction of arsenic acid to H_3AsO_3 . Hence write an overall equation for the reaction between iodide ions and arsenic acid.	
		Reduction: $2\text{H}^+ + \text{H}_3\text{AsO}_4 + 2\text{e}^- \rightarrow \text{H}_3\text{AsO}_3 + \text{H}_2\text{O}$ [1] Oxidation: $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$ Overall: $2\text{H}^+ + \text{H}_3\text{AsO}_4 + 2\text{I}^- \rightarrow \text{H}_3\text{AsO}_3 + \text{H}_2\text{O} + \text{I}_2$ [1]	
	(iii)	The iodine, I_2 , that is liberated from the reaction between arsenic acid and iodide ions can be estimated by titration against a standard thiosulfate, $\text{S}_2\text{O}_3^{2-}$, solution. $\text{I}_2 + 2\text{S}_2\text{O}_3^{2-} \rightarrow 2\text{I}^- + \text{S}_4\text{O}_6^{2-}$ When a sample of 25.0 cm^3 of arsenic acid reacts with iodide ions, it was found that 23.30 cm^3 of $0.200 \text{ mol dm}^{-3}$ solution of thiosulfate ions was required to react completely with the iodine liberated. Calculate the concentration of the sample of arsenic acid.	
		Amt of thiosulfate reacted = $\frac{23.3}{1000} \times 0.2 = 0.00466 \text{ mol}$ $\text{I}_2 \equiv 2\text{S}_2\text{O}_3^{2-}$ Amt of $\text{I}_2 = 0.00466 \div 2 = 0.00233 \text{ mol}$ [1] $\text{I}_2 \equiv \text{H}_3\text{AsO}_4$ Amt of H_3AsO_4 in $25.0 \text{ cm}^3 = 0.00233 \text{ mol}$ $[\text{H}_3\text{AsO}_4] = \frac{1000}{25} \times 0.00233 = 0.0932 \text{ mol dm}^{-3}$ [1]	
			[5]

- (c) The following graph shows some data on consecutive elements from period 2 and period 3.



F is known to react with C to form an ionic compound.
G is a better conductor of electricity as compared to F.
The chloride of F gives a slightly acidic solution when dissolved in water.

- (i) From the information given, identify F.
Hence, describe the type of bonding in F either in words or with a labelled diagram.

From the data given, 2nd I.E. of F is the lowest thus it must be a group II element in period 3.

Therefore F is magnesium. [1]

F has giant metallic structure with electrostatic forces of attraction between the cations and sea of delocalised electrons. [1] or diagram

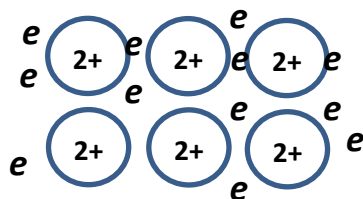


Diagram – [1]

- (ii) Suggest if the compound formed from F and C is soluble in water. You must support your answer with relevant explanations to gain full credit of this question.

Since the product is an ionic compound
There is favourable ion-dipole interactions which results in the release of energy that can break the giant ionic structure. [1]
Hence it is soluble in water. [1]

[4]

Total: [9]

2	(a)	For the reaction, $\text{CH}_3\text{CH}_2\text{CO}_2\text{H} (l) + \text{CH}_3\text{OH} (l) \rightleftharpoons \text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3 (l) + \text{H}_2\text{O} (l)$ The value of the equilibrium constant is 5.0.	For Examiner's use
	(i)	Suggest one way to increase the yield of the above reaction.	
		<u>Increase the amount of either reactants</u> (acid or alcohol) or <u>remove the products</u> (ester or water) as they are formed. [1]	
	(ii)	Explain, in terms of chemical bonding and structure, why the product, $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$ has low solubility in water.	
		<u>Weaker van der Waal's forces between solute</u> molecules are <u>not strong enough</u> to <u>displace</u> the <u>stronger hydrogen bonding between water</u> molecules for hydration to occur [1]	
	(iii)	State the numerical value of the equilibrium constant for the reverse reaction.	
		0.2 [1]	
	(iv)	Suggest the reagent and condition for the reverse reaction.	
		Reagent: dilute HCl or H_2SO_4 Condition: Heat } [1]	
	(v)	In an experiment, 1 mol of $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$ and 1 mol of water are mixed in $V \text{ cm}^3$. Calculate the amount of acid present at equilibrium.	

	$\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$	+ H_2O	$\rightleftharpoons$	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$	+ CH_3OH
Initial amt/ mol	1	1		0	0
Change/ mol	-x	-x		x	x
Eqm/ mol	1-x	1-x		x	x

Let x be amount of acid at equilibrium
 $K_c = 0.2$

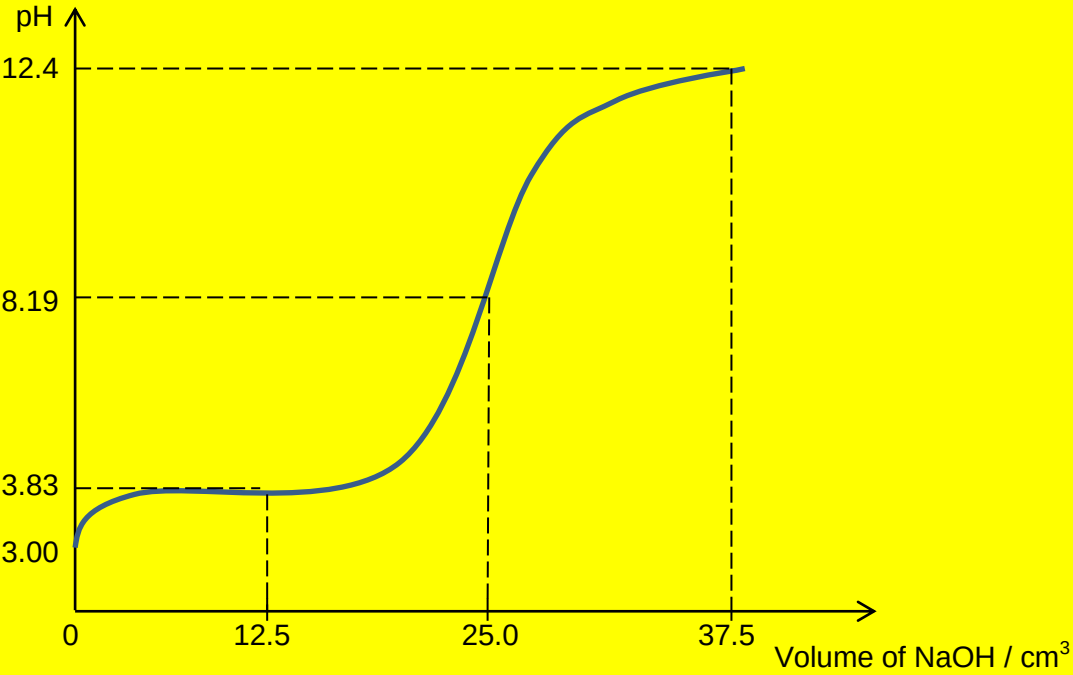
$$\frac{\left(\frac{x}{V}\right)\left(\frac{x}{V}\right)}{\left(\frac{1-x}{V}\right)\left(\frac{1-x}{V}\right)} = 0.2 \quad \text{[1] for showing V in working}$$

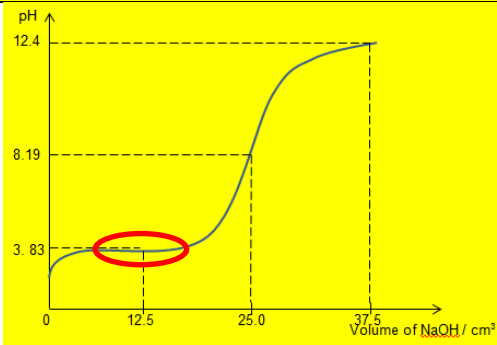
$$\frac{(x)}{(1-x)} = \sqrt{0.2}$$

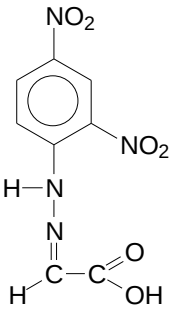
$x = 0.309 \text{ mol}$ [1] award ecf if K_c is calculated wrongly

		(vi)	Suggest a chemical test to distinguish between $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$.	
			Chemical test: Add <u>H_2SO_4 (aq), heat</u> followed by <u>KMnO_4</u> to separate test tubes of $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$. [1] Observation: $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$: <u>Decolourisation of purple</u> KMnO_4, <u>effervescence of CO_2</u> which forms white ppt when bubbled into $\text{Ca}(\text{OH})_2$ (aq). $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$: Decolourisation of purple KMnO_4, <u>no effervescence</u> [1] OR Test: Add <u>NaOH (aq)</u> and <u>heat</u> to each compound separately, followed by <u>I_2 (aq)</u>, NaOH (aq), heat. alternative[1] Observation: $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$: <u>No decolourisation of brown I_2</u> and <u>yellow ppt</u> CHI_3 $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$: <u>Decolourisation of brown I_2</u> and <u>yellow ppt</u> CHI_3 alternative [1]	
			Total: [8]	

3	Glycolic acid ($pK_a = 3.83$), $C_2H_4O_3$, is a colourless and odourless crystalline solid that is highly soluble in water. It is the smallest α-hydroxy acid (AHA) and is commonly used in skin care products and most often as a chemical peel performed by plastic surgeons due to its ability to penetrate the skin effectively. It can be prepared by the reaction of chloroethanoic acid ($pK_a = 2.85$) with hot sodium hydroxide followed by re-acidification.	For Examiner's use
(a)	(i) Write a chemical equation for the net reaction that has occurred showing the displayed formulae of all organic compounds.	
	$ \begin{array}{c} \text{Cl} \\ \\ \text{H}-\text{C}-\text{C} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{O}-\text{H} \end{array} \\ \\ \text{H} \end{array} + \text{NaOH} \longrightarrow \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{C} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{O}-\text{H} \end{array} \\ \\ \text{H} \end{array} + \text{NaCl} $ [1]	
	(ii) Explain why chloroethanoic acid has a lower pK_a as compared to glycolic acid.	
	In chloroethanoic acid there is presence of a stronger electron withdrawing Cl. This means the negative charge on the carboxylate ion of chloroethanoic acid will disperse to a greater extent thus stabilising the carboxylate ion of chloroethanoic acid. This facilitate the release of H^+ suggesting chloroethanoic acid to be a stronger acid to glycolic acid. [2]	
	(iii) Chloroethanoic acid is defined as a <i>weak acid</i>. With the aid of an equation, explain the words in <i>italics</i>.	
	A weak acid is one which dissociates partially in solution to give protons (H^+) [1] $\text{ClCH}_2\text{COOH} \rightleftharpoons \text{ClCH}_2\text{COO}^- + \text{H}^+$ [1]	

	(iv)	Beside pK_a, percentage dissociation is another way to measure the strength of an acid. Percentage dissociation is the ratio of the concentration of hydrogen ions to the concentration of the acid expressed in percentage. An analysis was carried out on a 25 cm^3 sample of a skin care product containing glycolic acid by titrating with 0.125 mol dm^{-3} of sodium hydroxide. The graph obtained is shown below. With this graph, determine the percentage dissociation of glycolic acid. 	For Examiner's use
		From graph $[H^+] = 10^{-3.00} = 1.00 \times 10^{-3}\text{ mol dm}^{-3}$ [1] Glycolic acid $\equiv$ NaOH [1] Amount of glycolic acid = Amount of NaOH = $25/1000 \times 0.125 = 0.003125\text{ mol}$ $[Glycolic\ acid] = 0.003125 / (25/1000) = 0.125\text{ mol dm}^{-3}$ [1] $\% \text{ dissociation} = [1.00 \times 10^{-3} / 0.125] \times 100\% = 0.8\%$ [1]	
	(v)	Prove that the final pH is 12.4 after 37.5 cm^3 of sodium hydroxide is added.	
		Vol of excess NaOH = $37.5 - 25 = 12.5\text{ cm}^3$ Amt of excess NaOH = $(12.5/1000) \times 0.125 = 1.56 \times 10^{-3}\text{ mol}$ [1] $[NaOH] = 1.56 \times 10^{-3} \div [(37.5+25)/1000] = 0.02496\text{ mol dm}^{-3}$ [1] $pOH = -\lg 0.02496 = 1.603$ $pH = 14 - 1.603 = 12.4$ (proven) [1]	

	(vi)	On the graph in (a)(iv), indicate the region where there is a buffer solution. State the species involved in this buffer system.	For Examiner's use
		 [1] Species present: $\begin{array}{c} \text{OH} \\ \\ \text{H}-\text{C}-\text{C} \\ \quad \diagup \quad \diagdown \\ \text{H} \quad \text{O} \quad \text{O}-\text{H} \end{array}$ & $\begin{array}{c} \text{OH} \\ \\ \text{H}-\text{C}-\text{C} \\ \quad \diagup \quad \diagdown \\ \text{H} \quad \text{O}^- \quad \text{Na}^+ \end{array}$ [1] for both	
	(vii)	Using an equation only, suggest how the pH remains fairly constant when small amount of sodium hydroxide is added to the solution in (a)(vi).	
		$\begin{array}{c} \text{OH} \\ \\ \text{H}-\text{C}-\text{C} \\ \quad \diagup \quad \diagdown \\ \text{H} \quad \text{O} \quad \text{O}-\text{H} \end{array} + \text{OH}^- \longrightarrow \begin{array}{c} \text{OH} \\ \\ \text{H}-\text{C}-\text{C} \\ \quad \diagup \quad \diagdown \\ \text{H} \quad \text{O}^- \end{array} + \text{H}_2\text{O}$ [1]	
		[13]	
(b)		The reaction scheme shows reactions involving glycolic acid. chloroethanoic acid $\xrightarrow{\text{I}}$ glycolic acid $\xrightarrow{\text{II}}$ $\text{C}_2\text{H}_2\text{O}_3$ $\xrightarrow{2,4\text{-DNPH}}$ Compound H $\downarrow \text{III}$ $\text{N} \equiv \text{C}-\text{CH}(\text{OH})-\text{COOH}$ Compound J $\xleftarrow{\text{PCl}_3}$ Compound I $\xleftarrow{\text{H}^+/\text{heat}}$ $\text{N} \equiv \text{C}-\text{CH}(\text{OH})-\text{COOH}$ $\xrightarrow{\text{LiAlH}_4}$ Compound K	
	(i)	State the reagents and conditions for reactions II and III. Reaction II: _____ Reaction III: _____	
		Reaction II: $\text{K}_2\text{Cr}_2\text{O}_7$, dilute H_2SO_4, distillation [1] Reaction III: HCN, NaCN (or NaOH), 10°C to 20°C [1]	

	(ii)	Draw the structural formulae for compounds H, I, J and K.	For Examiner's use
		HIJK	
		 H [1] $\begin{array}{c} \text{OH} \\ \\ \text{HOOC}-\text{C}-\text{COOH} \\ \\ \text{H} \end{array}$ I [1] $\begin{array}{c} \text{Cl} \\ \\ \text{ClOC}-\text{C}-\text{COCl} \\ \\ \text{H} \end{array}$ J [1] $\begin{array}{c} \text{H} \\ \\ \text{H}_2\text{N}-\text{CH}_2-\text{C}-\text{CH}_2\text{OH} \\ \\ \text{OH} \end{array}$ K [1]	
	(iii)	With the aid of the <i>Data Booklet</i> , suggest what will happen to the rate of reaction for reaction I when chloroethanoic acid is replaced with bromoethanoic acid.	
		Bond energy: C-Br (280) < C-Cl(340) [1] with relevant quote from Data Booklet Energy required: C-Br < C-Cl or Ease of cleavage of bond: C-Br > C-Cl Rate of reaction: bromoethanoic acid > chloroethanoic acid [1]	
		[8]	
	(c)	$2\text{C (s)} + 2\text{H}_2\text{(g)} + \text{O}_2\text{(g)} \xrightarrow{\Delta H_1^\circ} \text{CH}_3\text{COOH (l)}$ $\swarrow \Delta H_2^\circ$ $2\text{CO}_2\text{(g)} + 2\text{H}_2\text{O (l)}$ $\searrow \Delta H_3^\circ$ $2\text{CO}_2\text{(g)} + 2\text{H}_2\text{O (l)}$ Using the following data and the energy cycle above, calculate the standard enthalpy change of combustion of ethanoic acid. $\Delta H_c^\circ \text{ carbon} = -393 \text{ kJ mol}^{-1}$ $\Delta H_c^\circ \text{ hydrogen} = -286 \text{ kJ mol}^{-1}$ $\Delta H_f^\circ \text{ ethanoic acid} = -487 \text{ kJ mol}^{-1}$	
		By Hess Law: $\Delta H_1^\circ + \Delta H_3^\circ = \Delta H_2^\circ$ [1] $-487 + \Delta H_c^\circ \text{ ethanoic acid} = 2(-393) + 2(-286)$	

		$\Delta H_c^\ominus$ ethanoic acid = - 871 kJ mol ⁻¹ [1]	
			[2]
		Total: [23]	

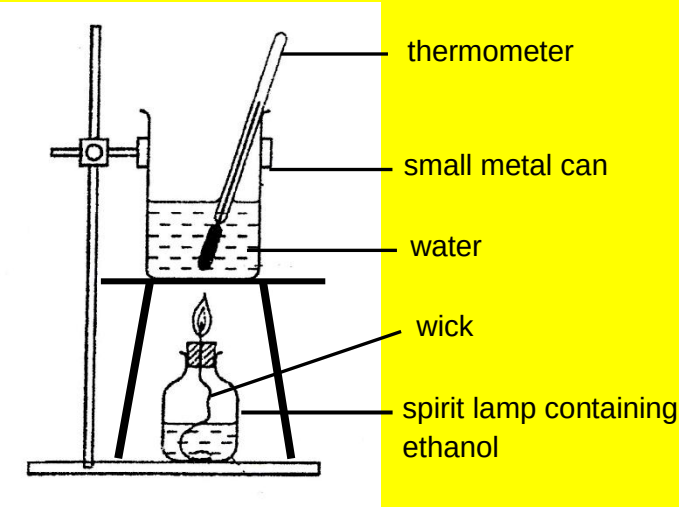
Section B

Answer two of the three questions in this section on separate paper.

4	(a)	Using sodium, phosphorous and sulfur, describe the reactions of their oxides with either sodium hydroxide or hydrochloric acid or both.	[4]																											
		The oxide changes from basic to acidic across the period. [1] Sodium oxide is a basic oxide, it undergoes neutralisation with acids to form salt and water. Na₂O(s) + 2 HCl(aq)→ 2 NaCl(aq) + H₂O(l) [1] Phosphorous oxides and sulfur oxides undergo neutralisation with bases to form salt and water P₄O₆(s) + 12 NaOH(aq) → 4 Na₃PO₃(aq) + 6 H₂O(l) or P₄O₁₀(s) + 12 NaOH(aq) → 4 Na₃PO₄(aq) + 6 H₂O(l) [1] SO₂(g) + 2 NaOH(aq) → Na₂SO₃ (aq) + H₂O(l) or SO₃(g) + 2 NaOH(aq) → Na₂SO₄(aq) + H₂O(l) [1]																												
	(b)	Propanone undergoes keto-enol tautomerism when reacted with iodine under an acidic medium at room temperature. The overall equation for this reaction is represented below: $\text{CH}_3\text{COCH}_3 + \text{I}_2 \xrightarrow{\text{H}^+} \text{CH}_3\text{COCH}_2\text{I} + \text{HI}$ To investigate how the rate of reactions depends on the concentration of each of the three reactants, two experiments were carried out. In each experiment the concentrations of two reactants were in excess and kept constant, whilst the concentration of the third reactant was measured over time. It is known that the order of reaction with respect to hydrogen ion is first order. <table><tr><td></td><td>reaction 1</td><td>reaction 2</td></tr><tr><td>time / s</td><td>[CH₃COCH₃] / mol dm⁻³</td><td>[I₂] / mol dm⁻³</td></tr><tr><td>0</td><td>1.50</td><td>1.50</td></tr><tr><td>30</td><td>0.96</td><td>1.28</td></tr><tr><td>60</td><td>0.62</td><td>1.05</td></tr><tr><td>90</td><td>0.38</td><td>0.83</td></tr><tr><td>120</td><td>0.25</td><td>0.60</td></tr><tr><td>150</td><td>0.15</td><td>0.38</td></tr><tr><td>180</td><td>0.09</td><td>0.15</td></tr></table>		reaction 1	reaction 2	time / s	[CH ₃ COCH ₃] / mol dm ⁻³	[I ₂] / mol dm ⁻³	0	1.50	1.50	30	0.96	1.28	60	0.62	1.05	90	0.38	0.83	120	0.25	0.60	150	0.15	0.38	180	0.09	0.15	
	reaction 1	reaction 2																												
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	(i)	Plot a graph of these results, putting all of the data on the same axes. Label each curve clearly.
		<u>Graph of [reactants]/mol dm⁻³ against time/s</u> axes and scale [1] Correct points plotted (randomly checked 3 points on each graph) [1] Straight line for I₂ and smooth curves for CH₃COCH₃ [1]
	(ii)	Use your graph to determine the order of reaction for each of the two reactants. Justify your answer in each case.
		I₂ is a downward sloping straight line, hence <u>rate is independent of the [I₂]</u>, therefore, <u>order of reaction wrt I₂ = 0</u> [1] CH₃COCH₃ show curve lines with <u>constant half-lives</u> of <u>46s</u>. [1] for clear annotation of two half-lives on graph Therefore, <u>order of reaction wrt CH₃COCH₃ = 1</u> [1]
	(iii)	Use your answer from (b)(ii) to write a rate equation for the reaction.
		<u>Rate = k [CH₃COCH₃][H⁺]</u> [1]
	(iv)	Explain how the rate of reaction would change if chlorine is used instead of iodine.
		Rate remained <u>constant</u> , since <u>rate is independent of the halogen</u> used / <u>order wrt halogen is zero</u> / <u>halogen is not involved in the rate determining step</u> . [1]

		(v)	With an aid of a suitable diagram, explain the effect on the rate of reaction if it is carried out in an ice bath.
			Fraction of molecules with energy, E Ice bath Room temperature Key: No. of particles with $E \geq E_a$ at higher temperature No. of particles with $E \geq E_a$ at lower temperature 0 E_a Energy, E Diagram [1] When temperature of the reaction decreases, • <u>average kinetic energy</u> of the reactant particles <u>decreases</u> • <u>less reactant particles with energy $\geq E_a$</u> <u>less effective collisions</u> Since rate of reaction is proportional to the frequency of effective collisions, <u>rate</u> of reaction <u>decreases</u>
		(vi)	Propanone can also react with iodine under basic medium resulting in a different organic product. Write an equation for this reaction, indicating the type of reaction involved and the observation.
			$\text{CH}_3\text{COCH}_3 + 3\text{I}_2 + 4\text{OH}^- \rightarrow \text{CH}_3\text{COO}^- + 3\text{I}^- + 3\text{H}_2\text{O} + \text{CHI}_3$ [1] <u>Oxidation</u> [1] <u>Decolourisation of brown</u> iodine solution and <u>pale yellow ppt</u> of CHI_3 formed. [1]
			[14]
	(d)		Propose a chemical test to differentiate CH_3COCH_3 from $\text{CH}_3\text{COCH}_2\text{I}$.
			Test: Add <u>NaOH (aq)</u>, <u>heat</u>, followed by <u>$\text{HNO}_3 \text{ (aq)}$</u> then <u>$\text{AgNO}_3 \text{ (aq)}$</u> [1] Observation: $\text{CH}_3\text{COCH}_2\text{I}$ will produce <u>yellow ppt</u> while <u>no ppt</u> is observed for CH_3COCH_3 [1]
			[2]
			Total: [20]

5	(a)	The table below shows the sources and the enthalpy changes of combustion of two common fuels used in vehicles. <table border="1" data-bbox="328 264 1350 405"> <thead> <tr> <th>Fuel</th><th>Main Sources</th><th>Enthalpy change of combustion / kJ mol^{-1}</th></tr> </thead> <tbody> <tr> <td>Octane</td><td>Refined from crude oil</td><td>– 5460.6</td></tr> <tr> <td>Ethanol</td><td>Fermentation</td><td>–1359.8</td></tr> </tbody> </table>	Fuel	Main Sources	Enthalpy change of combustion / kJ mol^{-1}	Octane	Refined from crude oil	– 5460.6	Ethanol	Fermentation	–1359.8
Fuel	Main Sources	Enthalpy change of combustion / kJ mol^{-1}									
Octane	Refined from crude oil	– 5460.6									
Ethanol	Fermentation	–1359.8									
	(i)	Define standard enthalpy change of combustion using ethanol as an example.									
		The standard enthalpy change of combustion of ethanol is the energy evolved when one mole of ethanol is completely burnt in excess oxygen at 298K and 1 atm pressure. [1]									
	(ii)	The figure below shows a diagram of a calorimeter used by a student to determine the enthalpy change of combustion of ethanol.  When 0.960 g of ethanol was combusted, the energy evolved heated 100 cm^3 of water from 25.0°C to 72.6°C. Using the information provided, determine the percentage efficiency, x, of this experiment, leaving your answer to three significant figures.									
		Heat absorbed by water, $Q' = 100 \times 4.18 \times (72.6 - 25.0)$ $= 19897 \text{ J}$ $= \mathbf{19.897 \text{ kJ [1]}}$ $n_{\text{ethanol burnt}} = \frac{0.96}{2 \times 12.0 + 6 \times 1.0 + 16.0} = 0.0209 \text{ mol}$ $\Delta H_c (\text{ethanol}) = -\frac{Q}{0.0209}$ $Q = 1359.8 \times 10^3 (0.0209) = 28.420 \times 10^3 \text{ [1]}$ $Q' = \frac{x}{100} Q$ $19.897 \times 10^3 = \frac{x}{100} \times 28.420 \times 10^3$ $x = \mathbf{70.0\% [1]}$									

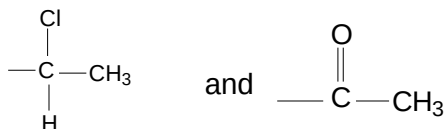
		(iii)	The fuel value (in kJ g^{-1}) of a substance is the heat energy released when 1 g of the substance is combusted. Calculate the fuel value for each of the two fuels. Hence, suggest a reason why octane has an advantage over ethanol as fuel for vehicles.
			Molar mass of octane = 114.0 g mol^{-1} Molar mass of ethanol = 46.0 g mol^{-1} Fuel value of octane = $5460.6 / 114 = 47.9 \text{ kJ g}^{-1}$ Fuel value of ethanol = $1359.8 / 46 = 29.6 \text{ kJ g}^{-1}$ [1] for both values Advantage: Octane has a larger fuel value. Thus it releases more energy per gram of fuel. For the vehicle to travel a specific distance, less amount of octane is needed as compared to using ethanol. [1]
		(iv)	Name the isomer of octane which has the lowest boiling point. 2,2,3,3-tetramethylbutane [1]
			[7]
	(b)		Ethanol has many uses in the organic laboratory. One of its common uses is to manufacture esters. $2\text{CH}_3\text{CH}_2\text{OH} + \text{HOOCCH}_2\text{COOH} \rightleftharpoons \text{CH}_3\text{CH}_2\text{OCOCH}_2\text{COOCH}_2\text{CH}_3 + 2\text{H}_2\text{O}$
		(i)	Deduce the effect on the equilibrium constant if a catalyst is introduced to the reaction chamber.
			When a catalyst is added, the rate of the forward and backward reactions will <u>increase by the same extent</u>. Therefore, the K_c value remain <u>unchanged</u>. [1]
		(ii)	Equal amounts of ethanol and propanedioic acid reacted together to reach equilibrium.

			Upon establishing equilibrium at time t , temperature of the reaction is decreased. Determine whether the forward reaction is exothermic or endothermic.
			By Le Chatelier's Principle, a decrease in temperature favours an <u>exothermic</u> reaction. When temperature is decreased, [reactants/ethanol] increases and [product/ester] decrease, indicating that the backward reaction is favoured . [1] Hence, the forward reaction is endothermic . [1]
		(iii)	State the observed changes in concentration of ethanol and ester as a result of the change that occurred at t_2 .
			Concentration of ethanol will decrease and concentration of ester will increase until a new equilibrium is reached. [1]
		(iv)	A student intern at the laboratory made the following claim: <i>For the esterification reaction above, the enthalpy change is known as the enthalpy change of neutralisation.</i> <i>However, the enthalpy change cannot be calculated accurately by using only the bond energies in the data booklet.</i> Comment on the intern's claim.
			The first sentence of the intern's claim is incorrect. Esterification reaction is not a neutralisation as the water produced in this case is not due to reaction between acid and base, it is a condensation reaction. [1] The second sentence is correct. The bond energies in the data booklet cannot be used directly to calculate enthalpy change as the reactants and products are not in the gaseous phase or $\Delta H_{\text{vaporisation}}$ need to be considered. [1]
			[6]
		(c)	An ester L, $\text{C}_8\text{H}_{13}\text{O}_2\text{Cl}$ undergoes reaction to give ethanol and acid M $\text{C}_6\text{H}_9\text{O}_2\text{Cl}$. Decolourisation occurs when 1 mole of ester L reacts with 1 mole of liquid bromine. When acid M reacts with hot acidified potassium manganate (VII), compound N is formed together with 2 moles of gas. Compound N is inert towards sodium carbonate powder and Fehling's reagent but produces orange crystals when reacted with hydrazine. 1 mole of compound N can react with alkaline aqueous iodine to give 1 mole of ethanedioate ions and 2 moles of yellow precipitate. Deduce the structural formulae of compounds L, M and N.
			[7]
			Ester L undergoes <u>acidic hydrolysis</u> [1] to form ethanol and acid M. Ester L undergoes <u>electrophilic addition</u> with liquid bromine, therefore ester L contains a <u>C=C double bond</u>. [1] Acid M undergoes <u>oxidation</u> with hot acidified potassium permanganate to give compound N and gas. Gas is CO_2. Therefore, acid M has a <u>terminal double bond or the oxidised product of HOOC-COOH</u>. [1] Compound N <u>does not undergo neutralisation with Na_2CO_3, absence of RCOOH</u>. [1]

Compound **N** does not undergo oxidation with Fehlings, absence of aldehyde. [1]

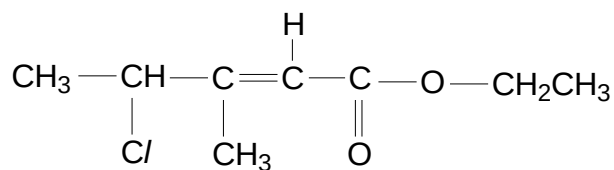
Compound **N** undergoes condensation with hydrazine, presence of ketone. [1]

1 mol of **N** undergoes nucleophilic substitution with NaOH to give alcohol [1]
followed by oxidation with alkaline iodine to give 2 mol of CHI₃, therefore **N** has:



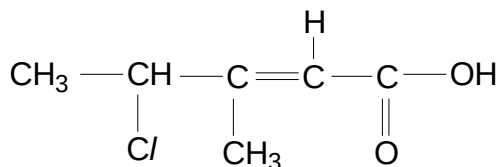
[1] Max – 4m for statements

L

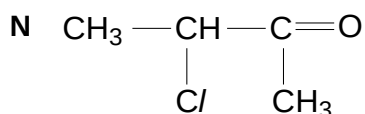


[1]

M



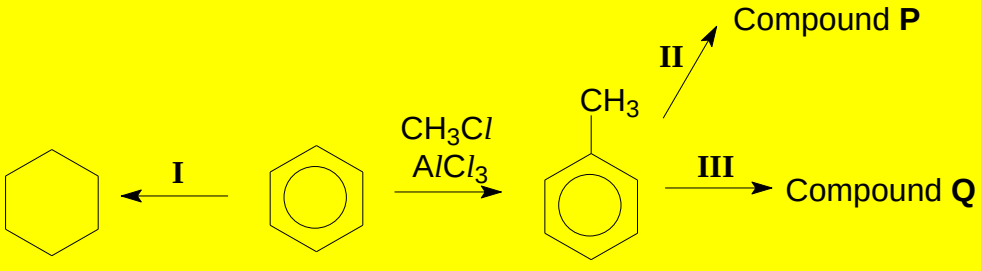
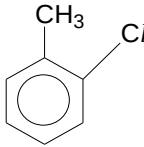
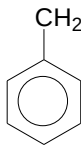
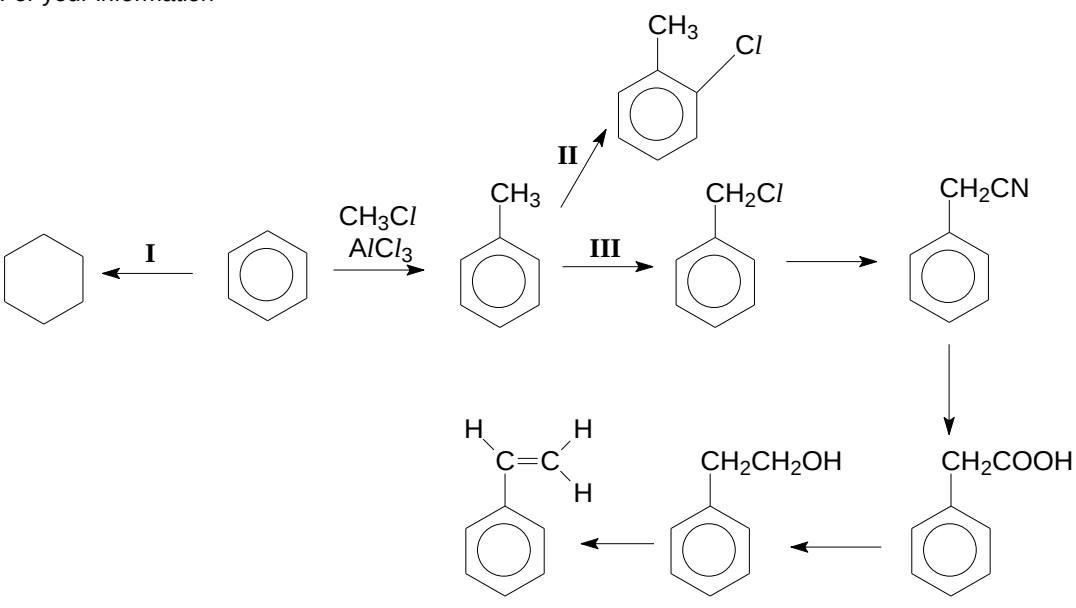
[1]



[1]

Total: [20]

6	(a)	(i)	The Period 3 elements vary in their melting points, electrical conductivities and pH of aqueous solutions of their chlorides. Sketch a clearly labelled graph to illustrate the pH of solutions of the highest chloride of the Period 3 elements (from Na to P). Correct axis & Correct shape – [1]
		(ii)	Describe what happens when aluminium chloride, AlCl_3, is dissolved in water, write an equation to illustrate this. AlCl_3 <u>undergo hydrolysis in water</u>, to form <u>an acidic solution</u>. [1] $\text{Al}(\text{H}_2\text{O})_6^{3+} + \text{H}_2\text{O} \rightleftharpoons [\text{Al}(\text{H}_2\text{O})_5(\text{OH})]^{2+} + \text{H}_3\text{O}^+$ [1]
		(iii)	Draw the structure of the compound formed when ammonia gas is pumped into a container with aluminium chloride. Hence, suggest the change in bond angle with respect to the aluminium atom in the structures before and after the reaction. [1] Lewis structure must be drawn. Bond angle around Al changes from 120° to 109.5° [1]
		(iv)	Boron trifluoride is also capable of similar reaction with ammonia. Explain why this is so. Boron has empty/ vacant / low lying 'p' orbital to accommodate the lone pair of electrons from nitrogen in the ammonia structure. [1]
		(v)	Boron trifluoride gas and aluminium chloride liquid are common catalysts used in isomerism, esterification and condensation reactions. Comment on the different physical states of the two catalysts. Both BF_3 and AlCl_3 have simple molecular structures. No of electron: $\text{AlCl}_3 > \text{BF}_3$ } [1] Extensiveness of VDW: $\text{AlCl}_3 > \text{BF}_3$ } Energy required to overcome VDW: $\text{AlCl}_3 > \text{BF}_3$ } [1] Boiling point: $\text{AlCl}_3 > \text{BF}_3$ } BF_3 is gas, AlCl_3 is liquid.
			[8]

(b)	(i)	The following reaction scheme shows some reactions involving benzene  Compound Q forms phenylethene in a 4-step synthesis reaction.
	(i)	State the type of reaction that took place for the conversion of methylbenzene from benzene.
		Electrophilic substitution. [1]
	(ii)	State the type of reaction that took place in reaction I.
		Reduction [1]
	(iii)	From the information given in the reaction scheme, draw the structures of compound P and Q which are isomers and propose the reagents and conditions required for reaction II and III. Compounds P and Q are isomers.
		Reaction II: AlCl_3, Cl_2, rtp [1] Reaction III: Cl_2, uv light [1]  Compound P:  [1] Compound Q:  [1] For your information 
		[6]

(c)		The ore, bauxite, consists of various forms of hydrated aluminium oxide, $Al_2O_3 \cdot nH_2O$. It is the raw material from which aluminium is obtained. The ore is usually found as a mixture of Al_2O_3 and Fe_2O_3. Some data on the properties of Al_2O_3 and Fe_2O_3 is tabulated below. <table border="1" data-bbox="491 369 1324 577"> <thead> <tr> <th></th><th>Al_2O_3</th><th>Fe_2O_3</th></tr> </thead> <tbody> <tr> <td>Melting point ($^{\circ}C$)</td><td>2072</td><td>1560</td></tr> <tr> <td>Solubility in hot concentrated sodium hydroxide</td><td>soluble</td><td>Insoluble</td></tr> <tr> <td>Molar mass ($g\ mol^{-1}$)</td><td>101.96</td><td>159.69</td></tr> </tbody> </table>		Al_2O_3	Fe_2O_3	Melting point ($^{\circ}C$)	2072	1560	Solubility in hot concentrated sodium hydroxide	soluble	Insoluble	Molar mass ($g\ mol^{-1}$)	101.96	159.69
	Al_2O_3	Fe_2O_3												
Melting point ($^{\circ}C$)	2072	1560												
Solubility in hot concentrated sodium hydroxide	soluble	Insoluble												
Molar mass ($g\ mol^{-1}$)	101.96	159.69												
	(i)	Suggest with the aid of an equation why aluminium oxide is soluble in hot concentrated sodium hydroxide whereas iron (III) oxide is not soluble.												
		Al_2O_3 is amphoteric, able to react as an acidic oxide, with the base, NaOH. Soluble salt formed. $Al_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaAl(OH)_4 (aq)$ [1] Fe_2O_3 metallic oxide and thus is basic, therefore it is unable to react with NaOH. [1]												
	(ii)	Draw the dot-and-cross diagram for aluminium oxide.												
		$2[Al]^{3+} \quad 3[\overset{\times}{\underset{\times}{\overset{\times}{\underset{\times}{O}}}}]^{2-}$ [1]												
	(iii)	With the aid of the <i>Data Booklet</i> , explain why aluminium oxide has a higher melting point as compared to iron (III) oxide.												
		Both aluminium oxide and iron (III) oxide have giant ionic structure. $\Delta H_{latt} \propto (q^+q^-)/(r^+ + r^-)$ r^+ of Al^{3+} (0.050) < r^+ of Fe^{3+} (0.064) [1] with relevant quote from <i>Data Booklet</i> $\Delta H_{latt} Al_2O_3 > \Delta H_{latt} Fe_2O_3$ More energy is required to overcome the electrostatic forces of attraction between the oppositely charged ions in Al_2O_3 resulting in higher melting point. } [1] (packing can be considered as an alternative answer).												
	(iv)	Suggest a reason why BeO would exhibit similar chemical properties as Al_2O_3 .												
		Be^{2+} and Al^{3+} have similar charge density [1] which results in diagonal relationship.												
		[6]												
		Total: [20]												

END